

MINCHUMINA, AK, USA

WMO: 702460

Lat: 63.886N

Lon: 152.302W

Elev: 207

StdP: 98.87

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26512

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.1	-35.9	-34.0	0.2	-27.4	-33.5	0.2	-26.7	6.9	-14.6	5.7	-16.4	0.0	30	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	25.3	15.1	23.0	14.2	22.0	13.8	16.4	22.9	15.4	21.3	14.5	19.9	1.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	10.0	18.0	12.7	9.4	17.1	12.1	9.0	16.9	46.5	23.0	43.7	21.2	40.9	19.9	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	4.8	3.9	-39.2	27.6	6.1	2.3	-43.6	29.2	-47.1	30.5	-50.5	31.8	-55.0	33.5		
			DB	WB												
					-34.8	17.6	1.7	1.4	-36.1	18.7	-37.1	19.5	-38.0	20.3	-39.3	21.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-19.8	-15.8	-11.5	-0.8	8.6	14.4	15.6	12.7	7.3	-3.2	-14.0	-18.3
	DBStd	14.12	8.75	7.28	6.44	5.60	4.34	3.29	3.02	3.23	3.80	5.58	6.36	7.10
	HDD10.0	4831	925	723	666	324	77	3	2	9	94	410	720	878
	HDD18.3	7443	1183	956	924	573	302	126	94	178	332	668	970	1136
	CDD10.0	446	0	0	0	1	33	134	173	93	12	0	0	0
	CDD18.3	18	0	0	0	0	0	7	8	3	0	0	0	0
	CDH23.3	164	0	0	0	0	7	64	72	21	0	0	0	0
	CDH26.7	18	0	0	0	0	0	9	7	2	0	0	0	0
Wind	WSAvg	0.9	0.7	0.7	1.0	1.1	1.3	1.2	1.3	1.0	0.8	0.8	0.6	0.6
Precipitation	PrecAvg	650	46	39	35	28	31	50	75	94	87	61	52	53
	PrecMax	853	89	95	78	69	62	99	132	167	167	174	122	106
	PrecMin	481	13	5	9	12	14	15	36	30	24	16	4	18
	PrecStd	99	22	19	17	13	13	18	24	26	42	34	32	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	1.3	5.4	14.8	24.0	27.8	27.6	26.1	18.8	11.1	2.2	-1.0
		MCWB	-1.9	-0.2	1.8	6.9	12.4	15.9	16.7	16.8	10.7	7.0	1.4	-2.4
	2%	DB	-3.8	-1.2	2.4	12.1	21.3	25.2	25.7	22.6	17.1	7.4	-0.2	-3.9
		MCWB	-4.9	-2.4	-0.5	5.2	11.2	14.5	15.6	15.2	10.1	4.6	-0.7	-4.8
	5%	DB	-6.5	-3.6	0.2	9.5	18.0	22.7	23.5	20.2	15.0	5.2	-2.7	-6.1
		MCWB	-7.1	-5.2	-1.9	4.0	9.5	13.6	14.9	14.2	9.4	3.1	-3.7	-6.6
	10%	DB	-8.7	-6.2	-2.2	7.5	16.4	21.1	22.1	17.9	12.9	3.0	-5.9	-8.7
		MCWB	-9.2	-7.2	-4.2	2.6	8.5	12.8	14.3	12.8	8.4	1.5	-6.7	-9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.0	-0.1	2.0	7.4	13.2	17.5	18.0	17.4	12.3	7.5	1.4	-2.1
		MCDB	-1.1	0.7	5.0	14.0	22.9	25.6	26.0	24.7	17.2	9.8	2.3	-1.1
	2%	WB	-4.4	-2.6	-0.1	5.3	11.7	15.7	16.5	15.8	11.2	5.0	-1.0	-4.4
		MCDB	-3.4	-1.5	2.0	10.9	19.8	22.7	23.4	21.2	15.6	7.1	-0.5	-3.6
	5%	WB	-6.8	-4.7	-1.9	4.2	10.4	14.6	15.6	14.8	10.2	3.4	-3.7	-6.3
		MCDB	-6.2	-3.2	0.1	9.2	17.0	21.2	21.7	19.2	14.0	5.0	-2.6	-5.9
	10%	WB	-8.8	-6.9	-4.0	2.8	9.3	13.5	14.9	13.7	9.3	2.0	-6.2	-9.0
		MCDB	-8.4	-5.8	-2.1	7.2	15.7	19.7	20.5	17.3	12.6	3.3	-5.6	-8.6
Mean Daily Temperature Range	5% DB	MDBR	6.6	8.2	10.0	10.1	10.4	9.6	8.6	7.8	7.9	5.4	6.3	6.9
		MCDBR	8.2	9.7	11.4	12.3	13.6	12.3	12.0	11.0	11.2	7.1	6.9	8.5
		MCWBR	7.6	8.7	8.9	7.4	6.6	5.7	5.2	5.3	6.3	4.9	6.6	8.0
	5% WB	MCDBR	8.2	9.5	10.6	11.7	12.7	10.9	10.6	9.6	9.7	6.7	6.8	8.2
		MCWBR	7.7	8.6	8.5	7.1	6.2	5.3	4.9	5.0	5.6	4.9	6.5	7.7
Clear-Sky Solar Irradiance	taub	0.184	0.246	0.255	0.292	0.338	0.371	0.367	0.388	0.299	0.264	0.186	0.176	
	taud	1.983	2.234	2.306	2.317	2.425	2.348	2.403	2.339	2.587	2.468	1.988	1.928	
	Ebn at Noon	430	700	859	892	875	846	839	777	813	690	404	187	
	Edh at Noon	27	55	78	98	98	109	100	98	62	44	25	12	
All-Sky Solar Radiation	RadAvg	0.21	0.93	2.75	4.66	5.23	5.42	4.65	3.55	2.31	0.95	0.29	0.07	
	RadStd	0.02	0.07	0.19	0.31	0.46	0.38	0.35	0.33	0.30	0.12	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+1.12	N/A	+0.71	N/A	N/A	N/A	-382	-435	+67	N/A

Nomenclature: See separate page